

Module 2

Probability distribution function:

Problems:

1. Given the following probability distribution, compute the value of 'a'. Also find $P(X > 6)$ and $P(X \leq 6)$.

X	1	2	3	4	5	6	7	8
P(X)	a	2a	3a	4a	5a	4a	3a	2a

Given that the distribution is a probability distribution

$$\therefore \sum_X P(X) = 1$$

$$\Rightarrow a + 2a + 3a + 4a + 5a + 4a + 3a + 2a = 1$$

$$\Rightarrow 24a = 1$$

$$\Rightarrow a = \frac{1}{24}$$

$$\begin{aligned} P(X > 6) &= P(x = 7 \text{ or } 8) \\ &= P(x = 7) + P(x = 8) \\ &= 3a + 2a = 5a = 5/24 \end{aligned}$$

$$\begin{aligned} P(X \leq 6) &= 1 - P(X > 6) \\ &= 1 - \frac{5}{24} = 19/24 \end{aligned}$$

2. If a random variable takes the values 1,2,3 and 4 such that $2P(x=1) = 3P(x=2) = P(x=3) = 5P(x=4)$, find the probability distribution of X.

Given that X assumes only 4 values 1,2,3 and 4.

Also $2P(x=1) = 3P(x=2) = P(x=3) = 5P(x=4)$,

$$\Rightarrow P(x=1) = \frac{1}{2} P(x=3)$$

$$P(x=2) = \frac{1}{3} P(x=3)$$

$$P(x=4) = \frac{1}{5} P(x=3)$$

Since the total probability is 1, we have

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$$P(x=1) + P(x=2) + P(x=3) + P(x=4) = 1$$

$$\Rightarrow \frac{1}{2}P(x=3) + \frac{1}{3}(x=3) + P(x=3) + \frac{1}{5}P(x=4) = 1$$

$$\Rightarrow P(x=3) \left[\frac{1}{2} + \frac{1}{3} + 1 + \frac{1}{5} \right] = 1$$

$$\Rightarrow P(x=3) = \frac{30}{61}$$

The probability distribution of X is

x	1	2	3	4
P(x)	30/122	30/183	30/61	30/305

Practice problem:

1. Given the following probability distribution, compute $P(|x| \leq 1)$, $P(|x| > 2)$ and

$$P(2x + 3 \leq 5).$$

x	-3	-2	-1	0	1	2	3
P(x)	1/6	1/6	1/6	1/12	1/12	1/6	1/6

Hint:

$$P(|x| \leq 1) = P(-1 \leq x \leq 1) = P(x = -1 \text{ or } x = 0 \text{ or } x = 1)$$

$$P(|x| > 2) = 1 - P(|x| \leq 2) = P(x = -2 + x = -1 + x = 0 + x = 1 + x = 2)$$

$$P(2x + 3 \leq 5) = P(2x \leq 5 - 3) = P(x \leq 1).$$

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